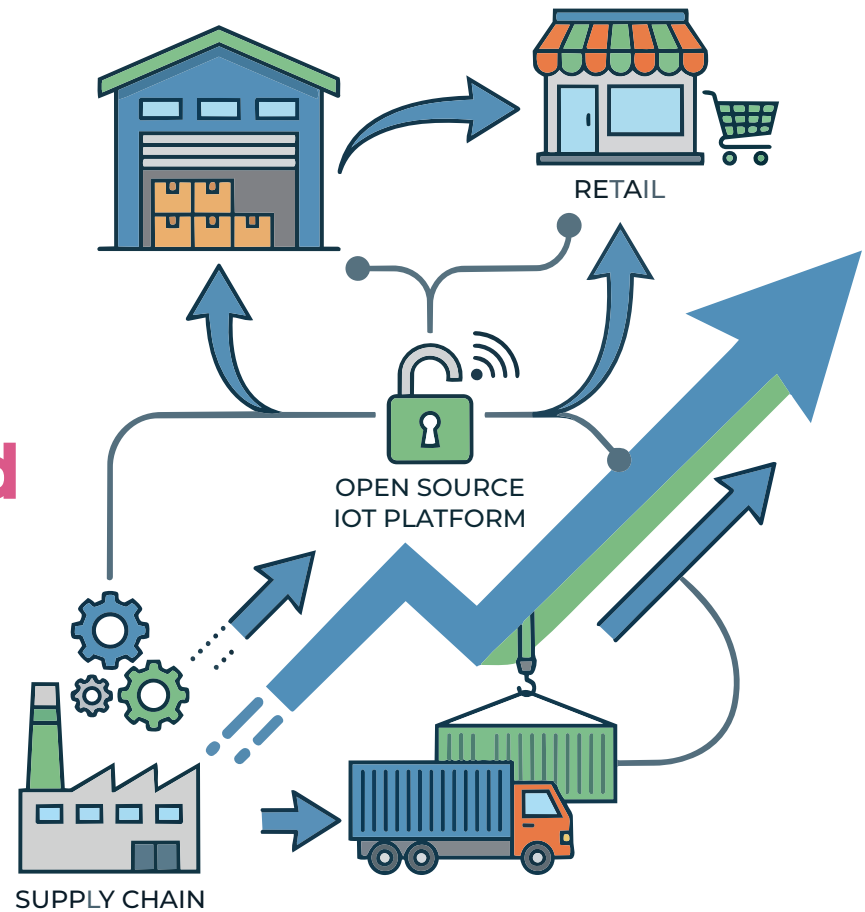
Scaling Open Source IoT Across Supply Chains and Retail

A field guide to the AI-native tool stack, the discipline that outlasts it, and the AI-quantum wave behind it.



In boardroom discussions around enterprise digital transformation, IoT is frequently mischaracterised as a hardware engineering problem. Executive teams invest millions in connected sensors, RFIDs, and smart gateways, only to watch initiatives stall at the pilot phase. Failure rarely stems from physical silicon; it arises because legacy architectures rely on proprietary, closed ecosystems that trap operational data in siloed cloud platforms, leading to exorbitant ingress fees and severe vendor lock-in.

To unlock measurable business value, CXOs must pivot from treating IoT as a tactical sensor deployment to viewing it as a core enterprise data fabric. Modern open source software stacks provide the strategic foundation needed to unify real-time physical events with enterprise resource planning (ERP) engines.

By shifting processing to the edge and streaming structured events into open data lakehouses, retail, consumer packaged goods (CPG), and supply chain organisations can convert raw telemetry into immediate bottomline impact.

The ERP-to-edge data bridge

The primary business challenge in modern supply chains is the time lag between physical reality on the warehouse floor and transactional records in ERP platforms such as SAP or Oracle.

Traditional architectures attempt to bridge this gap by shipping raw, unrefined telemetry directly to cloud databases, overwhelming data pipelines, and generating unsustainable cloud compute bills. The open source edge architecture resolves this by filtering, aggregating, and analysing telemetry at the physical boundary before pushing actionable signals directly into enterprise core systems.

The edge is also the AI layer

Most IoT investment cases stop at visibility — knowing where a pallet is, whether a shelf is empty, whether a compressor is drifting out of spec. But every layer of this architecture is also infrastructure for applied AI and treating it that way changes the ROI case.

At the edge, WasmEdge and ONNX Runtime aren't just lightweight containers — they're inference engines. Running a compact anomaly-detection or forecasting model directly on a retail gateway or a refrigerated transport unit means that a shrink event or a temperature excursion is flagged in